

Jason D. Hartline

Professor of Computer Science
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Research Interests

Economics. Mechanism design, auction theory, microeconomics, economic theory, econometrics.

Law. Algorithmic law, accountability, regulating algorithms.

Computer Science. Algorithmic mechanism design, algorithmic game theory, machine learning theory, algorithms, data science.

Education

Ph.D. in Computer Science. University of Washington, Seattle, WA. *Summer 2003*
Thesis: Optimization in the Private Value Model: Competitive Analysis Applied to Auction Design
Advisor: Anna Karlin.

M.S. in Computer Science. University of Washington, Seattle, WA. *Spring 2000*

B.S. in Computer Science. Cornell University, Ithaca, NY. *Spring 1997*

B.S. in Electrical Engineering. Cornell University, Ithaca, NY. *Spring 1997*

Current Appointment

Professor. Northwestern U., Evanston, IL. *Fall 2019 – present*
Department of Computer Science, McCormick School of Engineering;
Managerial Economics and Decision Sciences Department, Kellogg School of Management (courtesy);
and Department of Economics, Wienberg School of Arts and Sciences (courtesy).

Cofounder. Virtual Chair Inc., Chicago IL. *Summer 2020 – Current*

Previous Appointments

Visiting Professor. Stanford U., Stanford, CA. *2023-2024*
Economics department and Graduate School of Business

Associate Professor. Northwestern U., Evanston, IL. *Fall 2012 – Summer 2019*
Department of Computer Science, McCormick School of Engineering;
Managerial Economics and Decision Sciences Department, Kellogg School of Management (courtesy);
and Department of Economics, Wienberg School of Arts and Sciences (courtesy).

Visiting Researcher. Microsoft Research, Cambridge, MA. *Spring 2015*

Visiting Professor. Harvard U., Cambridge, MA. *2014*
Computer Science and Economics departments.

Assistant Professor. Northwestern U., Evanston, IL. *Winter 2008 – Summer 2012*
Electrical Engineering and Computer Science Department, McCormick School of Engineering and
Managerial Economics and Decision Sciences Department, Kellogg School of Management (courtesy).

Researcher. Microsoft Research, Mountain View, CA. 2004 – 2007
 Research Area: Algorithmic Mechanism Design, Auction Theory, Pricing Algorithms, Auctions for
 Sponsored Search.

Post-doctoral Research Fellow. ALADDIN, Carnegie Mellon U., Pittsburgh, PA. Fall 2003
 Research Area: Mechanism Design.
 Supervisor: Avrim Blum.

Mentoring

Ph.D. Advisees. current
 Anant Shah, Matthew vonAllmen, Chang Wang, Yifan Wu, Chenhao Zhang.

Former Students. since 2009
 Modibo Camara (Stanford), Yingkai Li (Yale), Yiding Feng (Microsoft Research), Bach Ha (Microsoft), Aleck Johnsen, Nima Haghpahan (Penn State, Asst. Prof.), Darrell Hoy (Carta), Michalis Mamakos (Northwestern, postdoc), and Samuel Taggart (Oberlin, Asst. Prof.)

Post-doctoral Fellows. since 2006
 Jinshuo Dong, Hedyeh Beyhaghi (CMU), Liad Blumrosen (Hebrew U.), Azarakhsh Malekian (Toronto)

Short-term Students. since 2004
 Gagan Aggarwal, Abraham Flaxman, Ning Chen, Mukund Sundararajan, Benjamin Prosnitz, Matthew Burgess, Saeed Alaei, Hu Fu, Shweta Jain, Rad Niazadeh, Sarah Lim, Sadie Hood.

Service

Program Chair. ACM Conference on Electronic Commerce. 2023

Program Committee. ACM Conference on Electronic Commerce. 2005, 2006, 2008 – present

Program Committee. ACM Symposium on CS and Law. 2023

Associate Editor. Journal of Economic Theory 2019 – present

Co-organizer. Northwestern CS+Econ Quarterly Workshop 2018 – present

Co-organizer. Northwestern Quarterly Theory Workshop 2016 – present

General Chair. ACM Conference on Electronic Commerce. 2020

Co-organizer. Special Quarter on Online Markets and Data Science Spring 2018
 with Jacob Abernethy, Constantinos Daskalakis, and Denis Nekipelov.

Special Initiatives Chair. ACM Special Interest Group on E-commerce. 2014 – 2015
 on the Academic Job Market.

Guest Editor. Games and Economic Behavior. 2011 – 2014
 special issues for papers from STOC, FOCS, and SODA conferences.

Advisory Editor. Games and Economic Behavior. 2012 – 2017

Associate Editor. Operations Research Letters. 2011 – 2014

Co-organizer. New York Computer science and Economics (NYCE) Day. 2013

Co-organizer. FOCS Workshop on Bayesian Mechanism Design.	2014
Program Committee. Symposium on Theory of Computation	2012
Co-organizer. Workshop on Bayesian Mechanism Design	2011
Co-organizer. Greece Economic and Algorithmic Theory Week.	2011, 2014
Co-organizer. Bertinoro Workshop on Algorithmic Game Theory.	2006, 2010
Tutorials Chair. ACM Conference on Electronic Commerce.	2010
Local Arrangements. ACM Conference on Electronic Commerce.	2008
Organizer. Midwest Theory Day.	2008
Program Committee. ACM-SIAM Symposium on Discrete Algorithms.	2007
Co-organizer. Bay Algorithmic Game Theory Symposium (biannual).	2006 – 2007
Co-organizer. Workshop on Sponsored Search Auctions.	2006
Co-organizer. Alternative Solution Concepts in Mechanism Design.	2006
Co-organizer. ALADDIN Workshop on Auction Theory & Practice.	2003

Awards, Fellowships, and Grants

FORC Best Student Paper. with Jiale Chen and Onno Zoeter for “Fair Grading of Randomized Exams”	2023
NSF AF. Mechanism Design for the Classroom	2022
ESA Test of Time. with Andrew Goldberg for “Competitive Auctions and Multiple Digital Goods” from ESA 2001.	2021
SIGecom Test of Time. with Andrew Goldberg and Andrew Wright for “Competitive Auctions and Digital Goods” from SODA 2021.	2021
NSF TRIPODS. Institute for Data, Econometrics, Algorithms and Learning with Aravindan Vijayaraghavan and others.	2019
NSF AitF. Mechanism Design and Machine Learning for Peer Grading with Douglas Downey and Eleanor O’Rourke.	2017
NSF AF. Non-revelation Mechanism Design	2016
Teacher of the Year. EECS Dept., Northwestern U.	2011
NSF ICES. Towards Realistic Mechanisms: statistics, inference, and approximation in simple Bayes-Nash implementation. With Shuchi Chawla and Denis Nekipelov.	2011
NSF CAREER Award. Mechanism Design	2009
NSF TF. Mechanism Design and Approximation with Shuchi Chawla.	2008

ALADDIN Post-doctoral Research Fellowship. Carnegie Mellon University.	2003
NSF Math Sciences Post-doctoral Research Fellowship. Declined.	2003
Microsoft Endowed Fellowship. CS Dept., U. of Washington	2001
Bob Bandes Teaching Award, Honorable Mention. CS Dept., U. of Washington	1998
Small Business Innovative Research Grant. Department of Education	1997

Patents

Online Pricing and Buyback. U.S. Patent #8260724 with Moshe Babaioff and Robert Kleinberg.	2012
Systems and Methods for Pricing and Selling Digital Goods. U.S. Patent #6985885 with Andrew Goldberg and Andrew Wright.	2006

Book Chapters

Profit Maximizing Mechanism Design. Algorithmic Game Theory with Anna Karlin; eds. Noam Nisan, Tim Roughgarden, Eva Tardos, and Vijay Vazirani.	2007
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Popular Press

Badminton and the Science of Rule Making. with Robert Kleinberg.	<i>Huffington Post</i> 2012
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Working Papers

ElicitationGPT: Text Elicitation Mechanisms via Language Models. with Yifan Wu.	2024
Subgame Optimal and Prior-Independent Online Algorithms. with Aleck Johnsen and Anant Shah.	2024
Algorithmic robust forecast aggregation. with Yongkang Guo, Zhihuan Huang, Yuqing Kong, Anant Shah, and Fang-Yi Yu.	2024
Mechanism Redesign. with Shuchi Chawla and Denis Nekipelov.	2017

Journal Papers

Fast Core Pricing for Rich Advertising Auctions. with Rad Niazadeh, Nicole Immorlica, Mohammad Resa Khani, and Brendan Lucier.	<i>Operations Research</i> 2022
When is pure bundling optimal?. with Nima Haghpanah	<i>Review of Economic Studies</i> 2021

- Bernoulli factories and black-box reductions in mechanism design.** *JACM* 2021
with Shaddin Dughmi, Robert Kleinberg, and Rad Niazadeh.
- Efficient Computation of Optimal Auctions via Reduced Forms.** *Math of OR* 2019
with Saeed Alaei, Hu Fu, Nima Haghpahan, and Azarakhsh Malekian.
- Optimal auctions vs. Anonymous Pricing.** *Games and Economic Behavior* 2018
with Saeed Alaei, Rad Niazadeh, Manolis Pountourakis, and Yang Yuan. Special issue.
- Non-optimal Mechanism Design.** *American Economic Review* 2015
with Brendan Lucier.
- Bayesian Incentive Compatibility and Matchings.** *Games and Economic Behavior* 2015
with Robert Kleinberg and Azarakhsh Malekian. Special issue.
- Mechanism Design via Consensus Estimates, Cross Checking, and Profit Extraction.**
with Bach Ha. Special issue. *Transactions on Economics and Computation* 2013
- Optimal Crowdsourcing Contests.** *Games and Economic Behavior* 2015
with Shuchi Chawla and Balu Sivan. Special issue.
- Envy freedom and prior-free mechanism design.** *Journal of Economic Theory* 2015
with Nikhil Devanur and Qiqi Yan. Special issue.
- Bayesian Mechanism Design.** *FTTCS*¹ 2012
- Approximation in Mechanism Design.** *American Economic Review* 2012
- Derandomization of Auctions.** *Games and Economic Behavior* 2010
with Gagan Aggarwal, Amos Fiat, Andrew Goldberg, Nicole Immorlica, and Madhu Sudan.
- Algorithms for Data Migration.** *Algorithmica* 2010
with Eric Anderson, Joseph Hall, M. Hobbess, Anna Karlin, Jared Saia, Ram Swaminathan, and John Wilkes.
- Reducing Mechanism Design to Algorithm Design via Machine Learning.** *JCSS*² 2008
with Maria-Florina Balcan, Avrim Blum, and Yishay Mansour.
- Competitive Auctions.** *Games and Economic Behavior* 2006
with Andrew Goldberg, Anna Karlin, Mike Saks, and Andrew Wright. Special issue.
- Characterizing History Independent Data Structures.** *Algorithmica* 2005
with Edwin Hong, Alexander Mohr, William Pentney, and Emily Rocke. Special issue.

¹Foundations and Trends in Theoretical Computer Science.

²Journal of Computer and Systems Sciences.

Refereed Conference Papers

- Fundamental Limits of Throughput and Availability: Applications to prophet inequalities and transaction fee mechanism design.** *EC*³ 2024
with Adityan Ganesh, Atanu Sinha, and Matthew vonAllmen
- A Decision Theoretic Framework for Measuring AI Reliance.** *FACCT*⁴ 2024
with Ziyang Guo, Yifan Wu, and Jessica Hullman.
- Designing Shared Info Displays for Agents of Varying Strategic Sophist.** *CSCW*⁵ 2024
with Dongping Zhang and Jessica Hullman.
- Error-Tolerant E-Discovery Protocols.** *CSLaw*⁶ 2024
with Jinshuo Dong, Liren Shan, and Aravindan Vijayaraghavan.
- Regulation of algorithmic collusion.** with Sheng Long and Chenhao Zhang *CSLaw* 2024
- Equivocal Blends: Prior Independent Lower Bounds.** *ITCS*⁷ 2024
with Aleck Johnsen
- The Rational Agent Benchmark for Data Visualization.** *VIS*⁸ 2023
with Yifan Wu, Ziyang Guo, Michalis Mamakos, Jessica Hullman
- Optimal Scoring Rules for Multi-dimensional Effort.** *COLT*⁹ 2023
with Liren Shan, Yingkai Li, and Yifan Wu
- Fair Grading Algorithms for Randomized Exams.** *FORC*¹⁰ 2023
with Jiale Chen and Onno Zoeter. Best student paper award.
- Screening with Disadvantaged Agents.** *FORC* 2023
with Hedyeh Beyhaghi, Modibo Camara, Aleck Johnsen, and Sheng Long
- Non-strategic Econometrics (for Initial Play).** *AAMAS*¹¹ 2023
with Daniel Chui and James Wright
- Simple Mechanisms for Non-linear Agents.** *SODA*¹² 2023
with Yiding Feng and Yingkai Li
- Algorithmic Learning Foundations for Common Law.** *CSLaw* 2022
with Dan Linna, Liren Shan, and Alex Tang
- Classification Protocols with Minimal Disclosure.** *CSLaw* 2022
with Jinshuo Dong and Aravindan Vijayaraghavan

³ACM Conference on Economics and Computation

⁴ACM Conference on Fairness, Accountability, and Transparency

⁵ACM Conference on Computer-Supported Cooperative Work and Social Computing

⁶ACM Symposium on Computer Science and Law

⁷Innovations in Theoretical Computer Science Conference

⁸IEEE Transactions on Visualization and Computer Graphics

⁹Conference on Learning Theory.

¹⁰Symposium on Foundations of Responsible Computing.

¹¹International Conference on Autonomous Agents and Multiagent Systems

¹²ACM-SIAM Symposium on Discrete Algorithms.

- Karp: A Language for NP Reductions.** *PLDI*¹³ 2022
with Chenhao Zhang and Christos Dimoulas
- Visualization Equilibrium.** *IEEE VIS*¹⁴ 2021
with Paula Kayongo, Glen Sun, and Jessica Hullman
- Revelation Gap for Pricing from Samples.** with Yiding Feng and Yingkai Li *STOC*¹⁵ 2021
- Welfare-maximizing Guaranteed Dashboard Mechanisms.** *EC* 2021
with Yuan Deng, Jieming Mao, and Balasubramanian Sivan
- Non-quasi-linear Agents in Quasi-linear Mechanisms.** *ITCS* 2021
with Moshe Babaioff, Richard Cole, Nicole Immorlica, and Brendan Lucier
- Mechanisms for a no-regret agent: Beyond the common prior.** *FOCS*¹⁶ 2020
with Modibo Camara and Aleck Johnsen
- Benchmark design and prior-independent optimization.** *FOCS* 2020
with Aleck Johnsen and Yingkai Li
- Inference from Prices.** *SODA* 2020
with Aleck Johnsen, Denis Nekipelov, and Zihe Wang.
- A Truthful Cardinal Mechanism for One-Sided Matching.** *SODA* 2020
with Rediet Abebe, Richard Cole, and Vasilis Gkatzelis.
- Sample Complexity for Non-Truthful Mechanisms.** *EC* 2019
with Samuel Taggart.
- Dashboard Mechanisms for Online Marketplaces.** *EC* 2019
with Aleck Johnsen, Denis Nekipelov, and Onno Zoeter.
- Optimal Auctions vs. Anonymous Pricing: Beyond Linear Utility.** *EC* 2019
with Yiding Feng and Yingkai Li.
- An End-to-end Argument in Mechanism Design (Prior-independent Auctions for Budgeted Agents).** With Yiding Feng. *FOCS* 2018
- Fast Core Pricing for Rich Advertising Auctions.** *EC* 2018
with Nicole Immorlica, Mohammad Reza Khani, Brendan Lucier, and Rad Niazadeh.
- Bernoulli Factories and Black-box Reductions in Mechanism Design.** *STOC* 2017
with Shaddin Dughmi, Robert Kleinberg, and Rad Niazadeh
- Bayesian Budget Feasibility with Posted Pricing.** *WWW*¹⁷ 2016
with Eric Balkanski.

¹³ACM SIGPLAN International Conference on Programming Language Design and Implementation

¹⁴IEEE Transactions on Visualization and Computer Graphics

¹⁵ACM Symposium on Theory of Computing.

¹⁶IEEE Symposium on Foundations of Computer Science.

¹⁷International Conference on the World Wide Web.

- A/B Testing of Auctions.** *EC 2016*
with Shuchi Chawla and Denis Nekipelov.
- No-regret Learning in Bayesian Games.** *NeurIPS¹⁸ 2015*
with Vasilis Syrgkanis and Eva Tardos.
- Reverse Mechanism Design.** *EC 2015*
with Nima Haghpanah.
- Optimal auctions vs. Anonymous Pricing.** *FOCS 2015*
with Saeed Alaei, Rad Niazadeh, Manolis Pountourakis, and Yang Yuan.
- Mechanism Design for Data Science.** *EC 2014*
with Shuchi Chawla and Denis Nekipelov.
- Price of Anarchy for Auction Revenue.** *EC 2014*
with Darrell Hoy and Samuel Taggart.
- Optimal Auctions for Correlated Buyers with Sampling.** *EC 2014*
with Hu Fu, Nima Haghpanah, and Robert Kleinberg.
- The Simple Economics of Approximately Optimal Auctions.** *FOCS 2013*
with Saeed Alaei, Hu Fu, and Nima Haghpanah.
- Auctions with Unique Equilibria.** *EC 2013*
with Shuchi Chawla.
- Prior-independent Auctions for Risk-averse Agents.** *EC 2013*
with Hu Fu and Darrell Hoy.
- Prior-free Auctions for Budgeted Agents.** *EC 2013*
with Nikhil Devanur and Bach Ha.
- Prior-independent Mechanisms for Scheduling.** *STOC 2013*
with Shuchi Chawla, David Malec, and Balu Sivan.
- Mechanism Design via Multi- to Single-agent Reduction.** *EC 2012*
with Saeed Alaei, Hu Fu, Nima Haghpanah, and Azarakhsh Malekian.
- Optimal Crowdsourcing Contests.** *SODA 2012*
with Shuchi Chawla and Balu Sivan. Invited to GEB special issue.
- Mechanism Design via Consensus Estimates, and Cross Checking, and Profit Extraction.** *SODA 2012*
With Bach Ha. Invited to TEAC special issue.
- Truth, Envy, and Profit.** *EC 2011*
with Qiqi Yan. Invited to JET special issue.
- Bayesian Incentive Compatibility and Matchings.** *SODA 2011*
with Robert Kleinberg and Azarakhsh Malekian. Invited to GEB special issue.

¹⁸Conference on Neural Information Processing Systems

- Bayesian Algorithmic Mechanism Design.** *STOC 2010*
with Brendan Lucier.
- Sequential Posted Pricing and Multi-parameter Mechanism Design.** *STOC 2010*
with Shuchi Chawla, David Malec, and Balasubramanian Sivan.
- Simple versus Optimal Mechanisms.** *EC 2009*
with Tim Roughgarden.
- Limited and Online Supply and the Bayesian Foundations of Prior-free Mechanism Design.** with Nikhil Devanur. *EC 2009*
- Selling Ad Campaigns: Online Algorithms with Cancellations.** *EC 2009*
with Moshe Babaioff and Robert Kleinberg.
- Mechanism Design and Money Burning.** *STOC 2008*
with Tim Roughgarden.
- Optimal Marketing Strategies over Social Networks.** *WWW 2008*
with Vahab Mirrokni and Mukund Sundararajan.
- Auctions for Structured Procurement.** *SODA 2008*
with Matthew Cary, Abraham Flaxman, and Anna Karlin.
- Algorithmic Pricing via Virtual Valuations.** *EC 2007*
with Shuchi Chawla and Robert Kleinberg.
- Knapsack Auctions.** *2006*
with Gagan Aggarwal.
- Bayesian Optimal No-deficit Mechanism Design.** *WINE¹⁹ 2006*
with Shuchi Chawla, R. Ravi, and Uday Rajan.
- Mechanism Design via Machine Learning.** *FOCS 2005*
with Maria-Florina Balcan, Avrim Blum, and Yishay Mansour.
- Derandomization of Auctions.** *STOC 2005*
with Gagan Aggarwal, Amos Fiat, Andrew Goldberg, Nicole Immorlica, and Madhu Sudan.
- On Profit-Maximizing Envy-Free Pricing.** *SODA 2005*
with Venkat Guruswami, Anna Karlin, David Kempe, Claire Kenyon, and Frank McSherry.
- Collusion-Resistant Mechanisms for Single Parameter Agents.** *SODA 2005*
with Andrew Goldberg.
- Near-Optimal Online Auctions.** *SODA 2005*
with Avrim Blum.
- From Optimal Limited to Unlimited Supply Auctions.** *EC 2005*
with Robert McGrew.

¹⁹Conference on Web and Internet Economics.

- On the Competitive Ratio of the Random Sampling Auction.** *WINE 2005*
with Uriel Feige, Abraham Flaxman, and Robert Kleinberg.
- Near-Optimal Pricing in Near-Linear Time.** *WADS²⁰ 2005*
with Vladlen Koltun.
- A Lower Bound on the Competitive Ratio of Truthful Auctions.** *STACS²¹ 2004*
with Andrew Goldberg, Anna Karlin, and Mike Saks.
- Competitiveness via Consensus.** *SODA 2003*
with Andrew Goldberg
- Envy-Free Auctions for Digital Goods.** *EC 2003*
with Andrew Goldberg.
- Truthful and Competitive Double Auctions.** *ESA²² 2002*
with Kaustubh Deshmukh, Andrew Goldberg, and Anna Karlin.
- Competitive Generalized Auctions.** *STOC 2002*
with Amos Fiat, Andrew Goldberg, and Anna Karlin.
- Characterizing History Independent Data Structures.** *ISAAC²³ 2002*
with Edwin Hong, Alexander Mohr, William Pentney, and Emily Rocke.
- Competitive Auctions and Digital Goods.** *SODA 2001*
with Andrew Goldberg and Andrew Wright.
- Competitive Auctions for Multiple Digital Goods.** *ESA 2001*
with Andrew Goldberg.
- On Algorithms for Efficient Data Migration.** *SODA 2001*
with Joe Hall, Anna Karlin, Jared Saia, and John Wilkes.
- An Experimental Study of Data Migration Algorithms.** *WAE²⁴ 2001*
with E. Anderson, J. Hall, M. Hobbess, A. Karlin, J. Saia, R. Swaminathan, and J. Wilkes.

²⁰Workshop on Algorithms and Data Structures.

²¹Symposium on Theoretical Aspects of Computer Science.

²²European Symposium on Algorithms.

²³International Symposium on Algorithms and Computation.

²⁴International Workshop on Algorithm Engineering.